Thurstone is the Model of Choice

Across eleven human datasets, a population's pairwise odds are less extreme than Luce renormalization predicts, and a Gaussian race predicts them better

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Abstract

Luce's axiom says that dropping options from a menu leaves the odds between any two survivors untouched, so a population's first choices already fix its head-to-head preferences. Thurstone's 1927 model disagrees: a favourite's edge counts for less in a short field, so head-to-head odds are less lopsided than the first choices imply. Across eleven human datasets, 148,000 responses, they are less lopsided every time. Gaussian contestant removal, which re-runs Thurstone's race without the dropped options, is an untuned alternative, calibrated to the first choices and nothing else. It is the closer prediction in all eleven datasets and, on held-out choices, closes a median 44 per cent of the gap between the axiom and the best rule the data could support. Luce's axiom is the special case with Gumbel noise, so adopting the general form forfeits nothing. The claim is about population shares, the assumption inside the Plackett–Luce likelihood, not about individuals.

1 Renormalization and the Gaussian contest

Suppose we know, for a population and a set S of alternatives, the share p_i choosing or first-ranking each item i. Luce's axiom [13] assigns each item a fixed worth $u(i)$ with

$$p_i = \frac{u(i)}{\sum_{j \in S} u(j)}, \tag{1}$$

so the worths are determined up to scale by the shares. The axiom then fixes what happens on any subset: the denominator changes and nothing else does, leaving p_i/p_j untouched for any two survivors. In particular the share preferring i to j when only those two are considered must equal $p_i/(p_i + p_j)$. This is Independence of Irrelevant Alternatives, and in the analysis of rankings it is the content of the Plackett–Luce likelihood.

Thurstone [20] links the same two quantities through a race. Item i has a location a_i and draws a performance $X_i \sim N(a_i, 1)$ on each occasion, and the item with the winning draw is chosen, so p_i is a winning probability in a field of $|S|$. The locations are recovered from the shares, and the pairwise margin then follows from a two-horse race between i and j alone. It is smaller in magnitude than the odds the shares suggest, because beating one rival is easier than beating several and a favourite's advantage counts for less when the field is short.

Figure 1 shows the two predictions on a four-item example. Both reproduce the four shares exactly, since each has enough freedom to do so. They disagree about the pairwise margin, and they disagree by a definite amount.

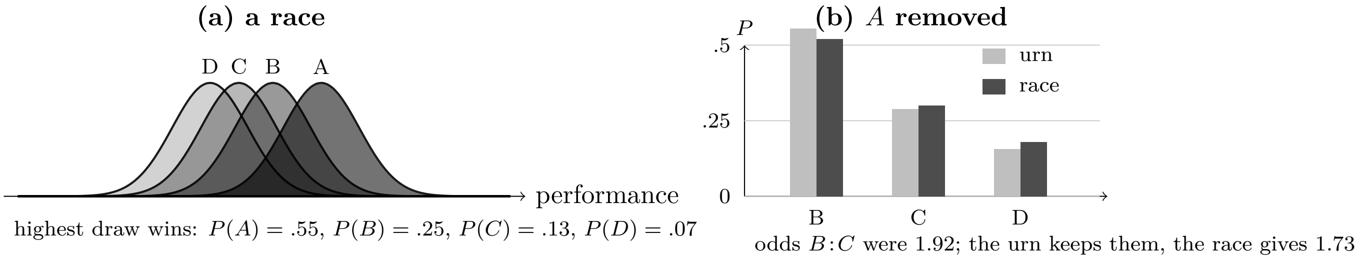

Figure 1: The two contenders disagree only under restriction. In (a) each item draws a performance and the highest wins. In (b) A is removed. The urn rescales, so every surviving odds ratio is unchanged by construction. The race is re-run without A and the vacated mass is redistributed unevenly, moving the $B : C$ odds from 1.92 to 1.73. Locations are calibrated to the probabilities in (a) by the lattice method of [4]; locations are recovered and the winning probabilities evaluated numerically, so the figures are exact to the precision shown.

Two claims follow, kept separate because they are logically independent and rest on different evidence, and because conflating them has been the main source of overstatement in this literature. Claim 1, that the nesting class is empirically necessary, is a statement about the Gumbel point: Yellott [25] showed that Luce's axiom holds within additive random utility models exactly when the performance noise is Type I extreme value, so renormalization is not a rival of the contest framework but a point inside it, and the surrounding class is required only because the data reject that point. The evidence is Section 5. Claim 2, that the untuned default in the class is already better, is a statement about Case V, the case Mosteller [18] made tractable and the one every textbook reaches for. It could have failed while Claim 1 held, since nothing obliges the closest member of a class to be the one chosen a century ago for its tractability. The evidence is Sections 5 and 5.1.

Neither prediction involves a quantity fitted to the pairwise data. Both calibrate $|S| - 1$ relative worths or locations from the shares alone, and then predict. We therefore compare zero-extra-parameter predictions, in the sense that the pairwise observations play no part in the calibration of either model.

Recovering locations from shares under Equation (1)'s competitor requires an integral. With independent equal-variance normals it is one-dimensional for any number of alternatives,

$$p_i = \int \phi(x - a_i) \prod_{k \neq i} \Phi(x - a_k)\, dx, \tag{2}$$

since conditioning on $X_i = x$ makes the other draws independent events. Once the locations are calibrated, the competing pairwise prediction is a single normal evaluation,

$$P(X_i > X_j) = \Phi\left(\frac{a_i - a_j}{\sqrt{2}}\right), \tag{3}$$

and locations are identified only up to a common translation, which we fix by $\sum_i a_i = 0$; the map from locations to strictly positive winning probabilities is one-to-one onto the interior of the simplex. What is genuinely high-dimensional is the correlated case, multinomial probit, and the likelihood of a complete ranking rather than of a top choice. The practical obstacle for most of the twentieth century was that Equation (2) must be evaluated numerically, and inverted numerically, once per candidate location vector inside an optimisation. A lattice method [4] now returns the whole vector of winning probabilities for fields of 10^5, which is what makes the inversion routine here.

2 Level of observation

Ten of the eleven datasets are complete rankings. From a ranking we can read off which alternative the respondent placed first, and also which of any two alternatives they placed higher. Both quantities are transformations of the same response, so the comparison in those ten datasets is between two accounts of one ranking distribution: the axiom says the pairwise margin is determined by the first-place shares in a particular way, and the race says it is determined differently.

That is not the same as observing choice when a menu is physically restricted. A respondent's ranking imposes a single ordering that applies to every subset by construction, so no ranking dataset can show a person preferring i to j in one menu and j to i in another, and none can reveal whether the person would in fact choose as their ranking implies when handed only two options. Any statement about menu restriction from these ten datasets rests on the assumption that they would. The eleventh dataset makes no such assumption, because subjects there chose separately from every subset.

The level of analysis matters as much as the elicitation. All eleven datasets give population quantities, and the aggregate failure of Independence of Irrelevant Alternatives does not imply that individuals violate it. A population of people who each choose by Equation (1), with worths differing from person to person, generally does not satisfy the axiom in aggregate, which is why mixed logit is a useful model and why its probabilities are integrals over heterogeneous logit probabilities rather than logit probabilities themselves [17]. So three claims must be kept apart. The first is that a single representative-agent Luce model, calibrated to aggregate first-place shares, predicts the aggregate pairwise margin. This paper rejects that claim. The second is that individuals have heterogeneous worths and each choose by a Luce rule, which would be consistent with everything reported here. The third is that a given individual, asked repeatedly, chooses by a Luce rule; nothing here addresses it, and the within-subject evidence in Section 7 speaks only to whether individual choices are order-consistent across menus.

3 Prior evidence

Independence of Irrelevant Alternatives has been contradicted repeatedly, and usually by manipulating the set. Debreu [5] pointed out that adding a close substitute should divide that alternative's support rather than draw proportionally from everything. Tversky [24] showed that similarity defeats simple scalability, of which the axiom is a special case. Huber, Payne and Puto [11] found that a dominated decoy raises the target's absolute share, which no random utility model allows, and Simonson [19] reported a related effect for compromise options. Luce [14] himself treated the empirical case as closed by 1977.

Econometrics kept the logit and relaxed the independence around it, through nested logit, the generalized extreme value family, and mixed logit [17], with a specification test from Hausman and McFadden [23]. The Thurstonian option was written down early and for exactly this purpose: Hausman and Wise [10] proposed multinomial probit as an estimator not constrained by the independence restriction, allowing both correlated performances and preferences that vary across people.

Neither response settles the question, and for the same reason. Both are flexible devices, and flexibility is the wrong instrument here. Mixed logit approximates any random utility model arbitrarily well [17], so a good mixed-logit fit is not evidence about the noise law inside the individual; the correlation, heteroskedasticity and taste-heterogeneity it might be absorbing are mutually confounded. Probit with a free covariance is in the same position, with the added difficulty that the covariance is identified only up to location and scale normalizations. A model fitted to the restriction discrepancy cannot be tested by it. Case V is the opposite kind of object: identity covariance, no free parameter, a point hypothesis that is able to lose.

The computational history has an irony worth recording. Probit was reported as practical for only a few alternatives, and simulation later made it estimable at cost [21], because a general-covariance choice probability is a $(J-1)$-dimensional orthant integral. That objection never applied to the independent case, where Equation (2) is one-dimensional and cheap. The contest class was set aside on a computational ground that does not bind on its most restrictive and most useful member.

Two results from this literature shape the present test. Yellott [25] showed that within additive random utility models with independent and identically distributed noise, and with menus rich enough to identify it, the axiom holds exactly when the noise is Type I extreme value, or Gumbel. The phrase double exponential is sometimes used for this law and sometimes for the Laplace; the Gumbel is meant. That result places the two accounts inside one framework, with their disagreement reduced to the shape of the noise, and it is why a single statistic can separate them. Block and Marschak [1] and Falmagne [6] characterised when a complete collection of choice probabilities admits any random utility representation, which Section 7 uses.

What has not been done, so far as we can establish, is to calibrate both accounts to the same aggregate shares and ask which predicts the pairwise margin. The classic demonstrations establish that an effect exists, on two- and three-item menus built from constructed profiles where there is no larger field to calibrate from. The econometric alternatives fit a covariance or a mixing distribution, so their success leaves open how much the unadorned race would have achieved. And where the two have been compared directly, the comparison was scored against no benchmark: a paired comparison of pair comparisons [8] concluded that the data conform somewhat more closely to Thurstone than to Luce but that the superiority is not definitive, which is the reading that has stood since. That verdict is a numerator without a denominator, of the kind Section 5.1 supplies.

4 Data

Eleven datasets have human respondents, complete rankings or ratings fine enough to yield one, and enough respondents for stable shares. Two rank-order card tasks from the General Social Survey ask what a child should learn and what matters in a job. Three independent samples of Jester users rate the same ten jokes on a continuous scale. Sushi rankings come from a Japanese web survey and film rankings are induced from Netflix ratings. There are perceptual discrimination tasks on dot arrays and puzzles, rankings of seven sports by participation preference, rankings of ten occupations by perceived prestige, and rankings of four political goals from a five-nation survey. The eleventh is the experiment of Costa-Gomes and colleagues [3], in which subjects chose from every subset of five consumer products; the archive identifies the products only by two-letter codes, so we describe them no more precisely than the archive does.

Counting is not straightforward and we report it plainly. The eleven rows comprise 147,719 dataset-responses. Unique people are fewer: the two General Social Survey tasks were fielded to overlapping samples of the same survey, while the three Jester files are disjoint user groups.

Two families of source were tried and set aside. Rating scales with eleven points cannot produce rankings of more than about five items, because respondents use only five or six of the levels available to them; in comparative election studies between 0.3 and 5 per cent of respondents give a tie-free ordering of nine parties. Preferential ballots fail for a different reason. Voters rank a few candidates and leave the rest unmarked, which the files record as a tie at the bottom, and reading that tie as indifference gives $\lambda = 0.68$ across fifty-three files where counting only pairs a voter actually ordered gives 0.15. Raw λ also varies widely across nominally identical elections: twelve contests for one body, all five-candidate, each with about ten thousand ballots, give values from 0.36 to 2.56, far wider than sampling error at those sample sizes. That spread is suggestive rather than conclusive, since λ under a fixed noise law depends on the share geometry, and we have not compared each election with its own model-predicted value. The tie-coding problem is sufficient on its own, and ballots are excluded on that ground.

5 Held-out predictive comparison

The comparison that decides the matter scores both accounts as probability forecasts on data neither has seen. Respondents are split into five folds. On the training folds we compute first-place shares over the full item set, and calibrate both accounts to those shares alone: the worths for renormalization are the shares themselves, and the race's locations come from inverting Equation (2). Each account then yields a predicted distribution on every subset of at least two items. For each held-out respondent and each subset, the outcome is that respondent's highest-ranked member of the subset, and we accumulate log loss. No pairwise or subset outcome enters either calibration.

Properness alone does not license reading a positive gain as evidence against the axiom. A proper score guarantees that the true probability vector minimises expected loss, not that a plug-in estimate of it beats a biased transformation of the same estimate. The shares handed to both accounts are estimated from a training fold, and the race contracts them, which is shrinkage toward equal subset probabilities. Shrinkage can lower held-out loss whether or not the population is non-Luce. Section **??** therefore replaces the comparison with zero by a comparison with the gain obtained when Luce is true by construction.

The race predicts held-out restricted choices better in every dataset, and every interval excludes zero. The margins are between 0.002 and 0.011 nats per prediction against losses near one nat. Read as effect sizes those numbers are negligible, and the next subsection argues that reading them as effect sizes is a mistake. Datasets are capped at five thousand respondents for runtime, and the occupational prestige rankings are omitted here because some alternatives are never ranked first in a training fold, leaving renormalization with a zero worth and no prediction to score.

5.1 The share of attainable improvement

Both accounts receive the same first-place shares, and those shares carry nearly all of the predictive content of a restricted menu. What a restriction map contributes is a thin residual on top, and its size is a property of the measurement rather than of the two hypotheses. Two facts

Dataset	n	K	Subsets	Renorm.	Race	Gain
Sushi	5,000	10	1,013	1.3727	1.3614	+0.0113 [+0.0096, +0.0130]
Political goals	2,262	4	11	0.8836	0.8767	+0.0069 [+0.0060, +0.0079]
GSS socialization	5,000	5	26	0.7986	0.7929	+0.0057 [+0.0043, +0.0072]
Sports participation	130	7	120	1.2594	1.2538	+0.0056 [+0.0021, +0.0094]
Jester file 3	5,000	10	1,013	1.5189	1.5165	+0.0024 [+0.0016, +0.0032]
Jester file 1	5,000	10	1,013	1.5287	1.5267	+0.0020 [+0.0013, +0.0027]
Jester file 2	5,000	10	1,013	1.5247	1.5230	+0.0018 [+0.0010, +0.0026]
GSS job values	5,000	5	26	0.8593	0.8575	+0.0018 [+0.0005, +0.0031]

Table 1: Mean held-out log loss per prediction, over every subset of size two or more, with both accounts calibrated only to training-fold first-place shares. Gain is the loss under renormalization minus the loss under the race, so positive favours the race; intervals are respondent bootstraps. The race is better in eight of eight datasets.

fix the scale. Renormalization and the race agree exactly whenever the shares within a subset are equal, by exchangeability, so every average over subsets includes cells where no discrepancy is possible. And the total available to any restriction map is bounded by what a perfect one would earn.

That bound is estimable. Alongside the two accounts we score a saturated forecast: the training-fold frequency with which each item is top within each subset, which is $|T| - 1$ free probabilities for a subset T. It uses the restricted data directly and is not available to a practitioner, but it is a legitimate out-of-sample forecast, and at this sample size it stands for the best a restriction map could do. The interesting quantity is then the fraction of attainable improvement,

$$\frac{\mathcal{L}_{\text{renorm}} - \mathcal{L}_{\text{race}}}{\mathcal{L}_{\text{renorm}} - \mathcal{L}_{\text{sat}}}, \tag{4}$$

which asks how far the race travels from renormalization toward that ceiling.

Dataset	Renorm.	Race	Saturated	Available	Captured
Jester file 1	1.5287	1.5267	1.5246	0.0042	48%
Jester file 3	1.5189	1.5165	1.5138	0.0051	47%
Political goals	0.8836	0.8767	0.8685	0.0150	46%
Jester file 2	1.5247	1.5230	1.5207	0.0040	44%
Sushi	1.3727	1.3614	1.3465	0.0262	43%
Sports participation	1.2594	1.2538	1.2401	0.0193	29%
GSS socialization	0.7986	0.7929	0.7761	0.0225	25%
GSS job values	0.8593	0.8575	0.8506	0.0088	20%

Table 2: Held-out log loss per prediction for the two share-only accounts and for a saturated restriction map fitted on the training folds. Available is the total any restriction map could earn on top of the shares; Captured is the race's share of it under Equation (4). Renormalization captures none of it by construction, since it is the unique map that adds no information.

The whole quantity in dispute is between 0.004 and 0.026 nats. A share-calibrated race, carrying $K - 1$ locations recovered from the $K - 1$ independent full-field shares and nothing fitted to the restricted data, captures a median 44 per cent of it, and between a fifth and a half in every dataset. The absolute margin was never an effect size; it was a numerator over a small ceiling, and the ratio is what bears on the generating process. Reporting the numerator alone is what makes a structural difference look like a rounding error.

Ordered outcomes are a natural place to look for a larger discrepancy, since each place would inherit the previous stage's error, and we report that we cannot yet do it. Pricing an exacta by removing the winner and re-running the race is a sequential rule of Harville's shape, and it is a coherent distribution over orderings, but it is not the race's own ordering law: for Gumbel noise sequential renormalization is exact, which is why Harville coincides with Plackett–Luce, and that coincidence is a property of the Gumbel rather than of sequential rules. Against the ordering law of the calibrated Gaussian race, obtained by simulation, the removal product misprices individual exacta and trifecta cells by up to a factor of 3.4 in a lopsided four-item field. That error exceeds the discrepancy between the two accounts it was meant to measure, so ordered-outcome pricing waits on the exact ordering law.

5.2 Contraction of the pairwise log odds

The direction of the difference is easier to see in a single summary statistic than in a log score. Let p be the first-place shares and q_{ij} the share placing i above j. The shift

$$\delta_{ij} = \log \frac{q_{ij}}{1 - q_{ij}} - \log \frac{p_i}{p_j} \tag{5}$$

is zero for every pair under renormalization, and negative under the race. The sign is a statement about oriented pairs: within each unordered pair we label i the higher-share alternative, since swapping the labels reverses δ. Fitting

$$\hat{\lambda} = \frac{\sum_{i<j} \log(p_i/p_j)\left(-\delta_{ij}\right)}{\sum_{i<j}\left[\log(p_i/p_j)\right]^2} \tag{6}$$

through the origin over unordered pairs, weighting each equally and dropping pairs where a log odds is undefined, gives the values in Table 3.

Dataset	Domain	Responses	$\hat{\lambda}$	Race predicts	Race error
GSS job values	values	21,797	0.227	0.234	0.007
Dots and puzzles	perception	6,363	0.188	0.161	0.027
Jester file 3	humour	24,904	0.374	0.337	0.037
GSS socialization	values	33,921	0.276	0.314	0.038
Jester file 1	humour	24,953	0.393	0.350	0.043
Sports participation	leisure	130	0.347	0.304	0.043
Jester file 2	humour	23,494	0.391	0.343	0.048
Sushi and Netflix	food, film	9,379	0.420	0.317	0.103
Consumer products	menu choice	373	0.419	0.231	0.188
Occupational prestige	social	143	0.591	0.300	0.291
Political goals	values	2,262	0.425	0.181	0.244

Table 3: Observed contraction and the race's prediction. Renormalization predicts zero, so its error equals the observed value in every row, from 0.188 to 0.591. Every row is a population-level quantity, including the consumer-products row, whose choice probabilities are pooled over subjects. Responses are dataset-responses; Section 4 discusses unique respondents.

Observed contraction is positive in all eleven rows, where representative-agent renormalization predicts zero. The race's error is under 0.05 in seven rows. Its residual has both signs: it under-predicts contraction in nine datasets and over-predicts in the two General Social Survey tasks, where observed values of 0.227 and 0.276 fall below predictions of 0.234 and 0.314. The three largest under-predictions come from the datasets with the fewest usable pairs, six, ten and fourteen, so we do not read a domain pattern from them.

The eleven rows are not eleven independent experiments. Three are Jester files, two are General Social Survey items, and several arrive through the same repositories, so the count describes the table rather than supporting a claim about independent replications. The held-out comparison in Table 1 is the basis for inference here, and its intervals are respondent bootstraps within dataset.

6 Contraction under other noise laws

Since the race under-predicts contraction more often than not, it is worth asking which noise laws contract more than a Gaussian, and the question has a clean answer that runs against intuition. Simulating populations whose members are races with a common noise law, standardised to unit variance, and computing the same λ:

Noise law	λ
Gumbel	0.001
Student t, 3 degrees of freedom	0.121
Student t, 8 degrees of freedom	0.203
Skew-normal, $a = 4$	0.078
Gaussian	0.245
Uniform on a bounded interval	0.352

Table 4: Contraction produced by different performance-noise laws in simulated races. The Gumbel row recovers the axiom, as Yellott's theorem requires, and also checks the pipeline. Heavier tails contract less than the Gaussian; bounded support contracts more.

Heavier tails move the model toward the axiom rather than away from it, so a heavy-tailed race is the wrong candidate for datasets that contract more than Gaussian. Bounded or thinner-tailed noise is the right direction there. Skewness also reduces contraction, which makes it a candidate for the two General Social Survey rows, where observed contraction falls short of the Gaussian prediction. Other explanations are not excluded by this exercise and are not distinguished by it, among them unequal variances across alternatives, correlated performances, heterogeneity across respondents, and utilities that depend on the set.

7 Within-subject choice from all subsets

The consumer-products experiment observes what the ranking datasets cannot. Its 373 subjects faced all thirty-one menus, one subject at a time, so a reversal is observable rather than excluded by construction, and the choices are individual rather than aggregate.

Reversals are uncommon. Of 26,099 comparisons in which an item chosen from a larger menu was still available in a smaller one, 829 switched away from it, a rate of 3.2 per cent; among subjects required to choose rather than permitted to defer, and so free of any coding decision, the rate is 3.9 per cent. The subject-level distribution is uneven. Six of the 373 record no usable choice, so 367 enter this count. Of those, 235 never reverse, the median subject reverses never, the ninetieth percentile reverses eight times, and the most violating tenth of subjects account for 58 per cent of all reversals. So most individuals are order-consistent across menus and a minority are markedly not, which is a pattern an average rate conceals.

Pooling subjects gives choice probabilities for all thirty-one menus, and the Block–Marschak sums test whether those probabilities admit any random utility representation. Of eighty computable sums, nine fall below zero and one below -0.02, the worst at -0.055; among forced-choice subjects sixteen fall below zero and ten below -0.02, the worst at -0.088. Estimated probabilities can produce small negative sums by sampling noise alone, and we have not computed the distance from the random utility polytope or intervals for the individual sums, so we report the counts and draw no strong conclusion. The complete-ranking datasets cannot contribute to this question at all, since a distribution over rankings is a random utility representation by construction.

8 Discussion

Adopting the class costs nothing in generality, which is the practical content of Claim 1. Because the axiom is nested rather than adjacent, a contest model reproduces renormalization whenever a population behaves that way, and Section 6 confirms it does so exactly, returning a contraction of 0.001 under Gumbel noise. There is no scenario in which choosing the class costs an analyst anything except the integral.

Claim 2 carries the stronger practical consequence, and it survives the objection that its margins are slight. Case V's advantage over renormalization is a fifth to a half of everything a restriction map could deliver, because the two accounts are constrained to agree at symmetric fields and the total on offer is small for that reason rather than because the two accounts are close. An analyst holding nothing but aggregate shares gets that fraction for free.

The boundaries of both claims should be stated as plainly as their content. What is rejected is a representative-agent Luce model of a population. Heterogeneous individuals who each choose by a Luce rule would generate aggregate violations of this kind, so nothing here shows that individual choice departs from logit, and the within-subject experiment speaks to order-consistency rather than to functional form. Ten of the eleven datasets compare two accounts of a ranking distribution rather than observing menu restriction, and their contribution should be read as such.

Case V is also not the end of the matter. Its residual has both signs, and Section 6 identifies the direction each sign points: bounded or thinner-tailed noise where contraction exceeds the Gaussian prediction, skewness or heavier tails where it falls short. Whether one noise law fits all eleven datasets is answerable with the same data and is not answered here.

Two limitations bound what has been shown. The political goals, occupational prestige and consumer datasets rest on six, ten and fourteen usable pairs, so their contraction estimates are the least secure in Table 3. And nothing here addresses regularity violations of the decoy and compromise kind, which lie outside both accounts.

For psychometrics the consequence concerns defaults. Plackett–Luce and Bradley–Terry likelihoods are standard for rankings and paired comparisons and they carry the axiom with them, largely because the contest was once impractical to compute. That reason has expired, the axiom is available as a nested special case, and on this evidence the ordering of defaults should be reversed for population-level ranking data.

Data and code

Sushi rankings come from Kamishima [12]; Jester ratings from the Eigentaste project [7], gauge-set files one to three; the General Social Survey cumulative file from NORC; the perceptual, leisure, prestige and political-goal rankings via PrefLib [15] and the CRAN packages BayesMallows, ConsRank, PerMallows and PLMIX; the menu-choice data from the replication archive of [3]. All are public and free. Analysis code, derived tables and cached copies are under `research/human` in `github.com/microprediction/winning`, together with the lattice calibration.

Two storage conventions must be told apart when reproducing this. BayesMallows, ConsRank and PerMallows store ranks, so element j holds the rank of alternative j, while PLMIX stores orderings, so the first element names the favourite. Reading one as the other transposes the data silently, and no error is raised. We established which is which by cross-validation: the political goals data ships in two of these packages, and the two agree only when each is read under its own convention.

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